

NLP 202 - Assignment I

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Abstract

This report presents the results of sentiment analysis using logistic regression on the IMDB dataset. The objective is to classify movie reviews as positive or negative. We compare baseline and tuned models, analyze hyperparameters, and present accuracy metrics.

1 Introduction

Sentiment analysis is a natural language processing (NLP) technique used to determine the sentiment conveyed in a text. In this report, we utilize logistic regression for binary classification of movie reviews from the IMDB dataset. Sentiment classification is crucial for various applications, including customer feedback analysis, opinion mining, and recommendation systems.

2 Dataset Description

The IMDB dataset consists of 50,000 movie reviews, equally split between positive and negative sentiments. Each review contains unstructured text data expressing a user's opinion about a movie. The dataset is preprocessed to remove stopwords, punctuation, and numerical characters before being used for model training.

For this study, the dataset is split into:

- 20,000 reviews for training (80%)
- 5,000 reviews for validation (20%)

The training set is shuffled to ensure diverse data representation during model training.

3 Baseline Model

The baseline model is a simple logistic regression classifier using word embeddings. The model uses a vocabulary size of 25,000, with each word represented as a dense vector of 100 dimensions. The classifier is optimized using the Adam optimizer with a learning rate of 0.001.

3.1 Baseline Model Results

The performance of the baseline model is evaluated over 10 epochs. The training and validation losses decrease over time, and the accuracy improves as shown below.

4 Hyperparameter Tuning

To improve model performance, we experimented with different hyperparameters, including:

- Learning rate: [0.001, 0.0005, 0.0001]
- Batch size: 32
- Training epochs: 10

The best results were obtained with a learning rate of 0.0001 with an accuracy of 0.89, which provided a balance between training stability and convergence speed.

5 Results and Performance

Figure 1 shows the accuracy trend over epochs. The tuned model achieved an accuracy of 86% on the validation set, demonstrating improved performance compared to the baseline.

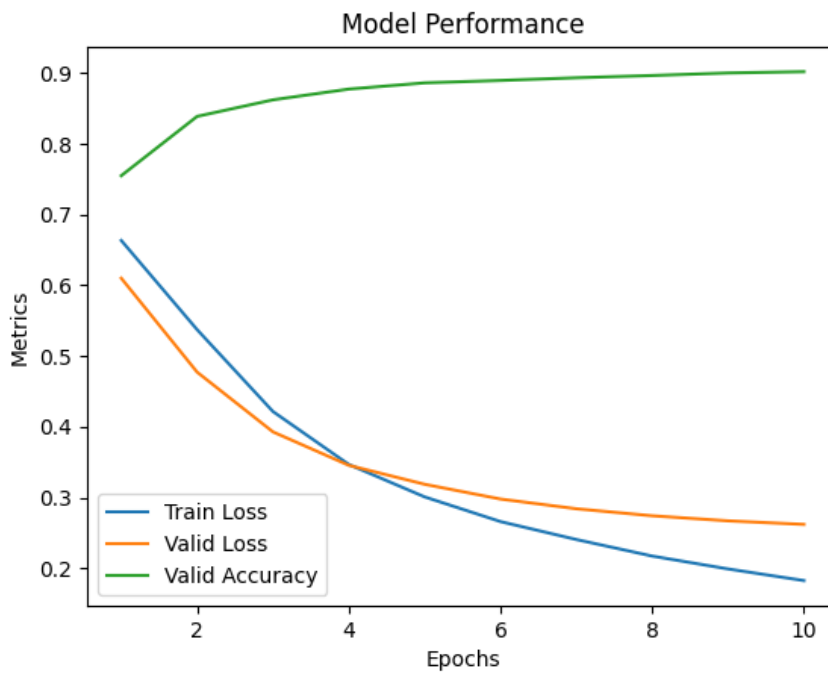


Figure 1: Validation Accuracy Over Training Epochs

6 Conclusion

This report demonstrates the effectiveness of logistic regression for sentiment analysis. The tuned model achieves an accuracy of 86% on the validation set, indicating significant improvement over the baseline. Future work could explore deep learning models such as LSTMs or Transformers for further enhancements.

Further Hyperparameter tuning and LSTM model will be submitted before the deadline.